California Environmental Protection Agency

AIR RESOURCES BOARD

QUALITY MANAGEMENT BRANCH

STANDARD OPERATING PROCEDURES

FOR

VERIFICATION AND CALIBRATION OF TEMPERATURE DEVICES

Standards Laboratory SOP 006

First Edition

MONITORING AND LABORATORY DIVISION

JULY 2014

California Environmental Protection Agency

AIR RESOURCES BOARD

Standard Operating Procedures (SOP) Approval

Title:	Verification and Calibration of Temperature Devices
SOP:	006, Revision 0
Section:	Data Analysis and Special Projects Section
Branch:	Quality Management Branch
Division:	Monitoring and Laboratory Division
Prepared by:	Robert Russell, Air Pollution Specialist
Approval:	This SOP has been reviewed and approved by:

Jeff Wright — 7/11/14

Jeff Wright, Manager — DATE

Data Analysis and Special Projects Section

Quality Management Branch

Jeff Wright for MM — 7/11/14

Michael Miguel, Chief — DATE

Quality Management Branch

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Verification and Calibration of Temperature Devices

1.0 Introduction

This procedure will provide National Institute of Standards and Technology (NIST) traceable verifications and calibrations for temperature devices in the range of 0.00 to 50.00 °C. Platinum Resistance Thermometers (PRT), Thermistors, and Thermocouple temperature sensors are acceptable. Probe dimension limitation is 1/8” to 1/2” diameter. All hand-held digital temperature meters are generally acceptable.

2.0 Regulatory Requirements

U.S. Environmental Protection Agency guidance document: Quality Assurance Handbook for Air Pollution Measurement Systems Volume IV: Meteorological Measurements, Version 2.0, describes the types of temperature measuring devices in the field environment, including the operation and maintenance, calibration, and auditing procedures. The recommended tolerance for a calibration or audit is 0.5 °C.

3.0 Summary of Method

Temperature probe devices are compared against a NIST traceable PRT at five points: 0, 20, 30, 40, and 50 °C. If the comparison meets the verification criteria (±0.5 °C at each point), a verification report is issued. No adjustments or corrections are needed before using the data from the display of the candidate device. If an adequate coefficient of determination (R2) is developed from comparing the candidate device to the reference standard (R2 ≥ 0.999), a calibration correction equation is also provided. A client may choose to use this calibration formula to correct the candidate’s display to improve the accuracy of the readings.

If the temperature device does not pass, a re-test is performed to confirm the failure. If the re-test confirms the failed criteria, the device must be repaired by the client before further testing is performed.

4.0 Summary of Changes to Previous Version

Not Applicable

5.0 Definitions

- Calibration – establishes a correction formula to be used to adjust or correct the display of the candidate instrument. This is determined through a comparison of the candidate instrument to a known, or reference standard.

- Verification – establishes comparability of a candidate instrument to a known, or reference standard; the output of candidate instrument is not corrected based upon the results of the verification procedure.

6.0 Personnel Qualifications

Before new personnel perform this procedure, one or more weeks of training from Standards Laboratory staff is required. Upon completion of this training, new personnel must to be able to demonstrate competency in performing this procedure without assistance.

7.0 Health and Safety

Because of the design of temperature probes, care should be taken when handling the probes to avoid injury**.**

8.0 Cautions

- **DO NOT** use pliers or other similar devices to squeeze the sheathing of the temperature sensor around the probe wiring harness. This could permanently damage the sensor and change the calibration coefficient.

- **DO NOT** use fluid to clean inside the well of the calibrator.

- It is important to keep the well of the calibrator clean and clear of any foreign matter.

9.0 Interferences

- Avoid operating the instrument in an oily, wet, or dirty environment.

- Operating the instrument in a draft-free environment improves performance of the instrument.

10.0 Equipment and Supplies

- Platinum Resistance Thermometer (PRT).

- Model 9142 Fluke Field Metrology Well (-25 to 150 °C).

11.0 Verification/Calibration Procedure

11.1 Power up the Fluke Metrology Well. See Figure 1.

Figure 1: Fluke Metrology Well Power Switch

11.2 The metrology well has six hole sizes. Insert the Fluke reference PRT into the hole that best matches the diameter of the probe. Insert the probe until it bottoms out; then raise it about one inch. See Figure 2.

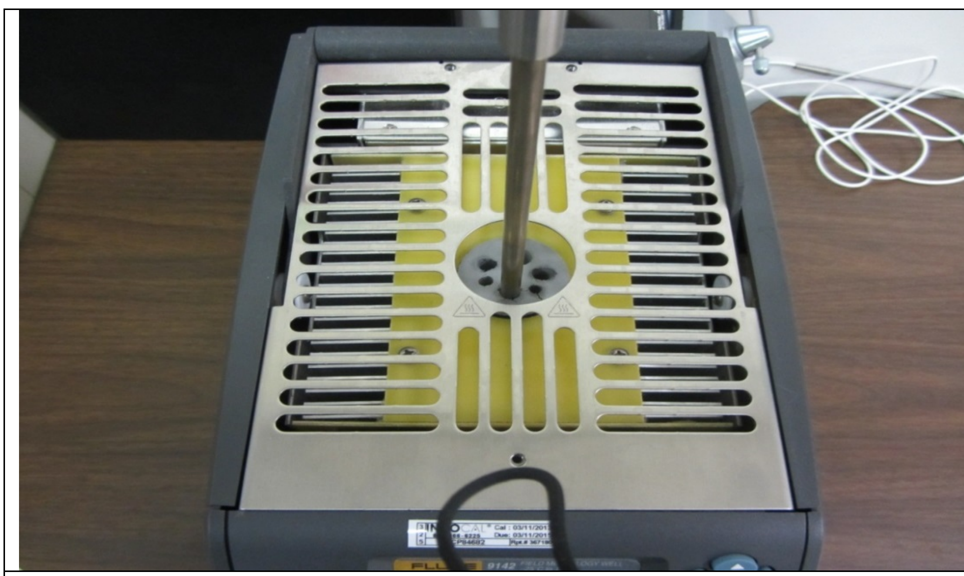


Figure 2: Reference Probe Inserted

11.3 Insert the candidate temperature probe into the hole that best matches the diameter of the probe. See Figure 3. Do not force the probe into the well.

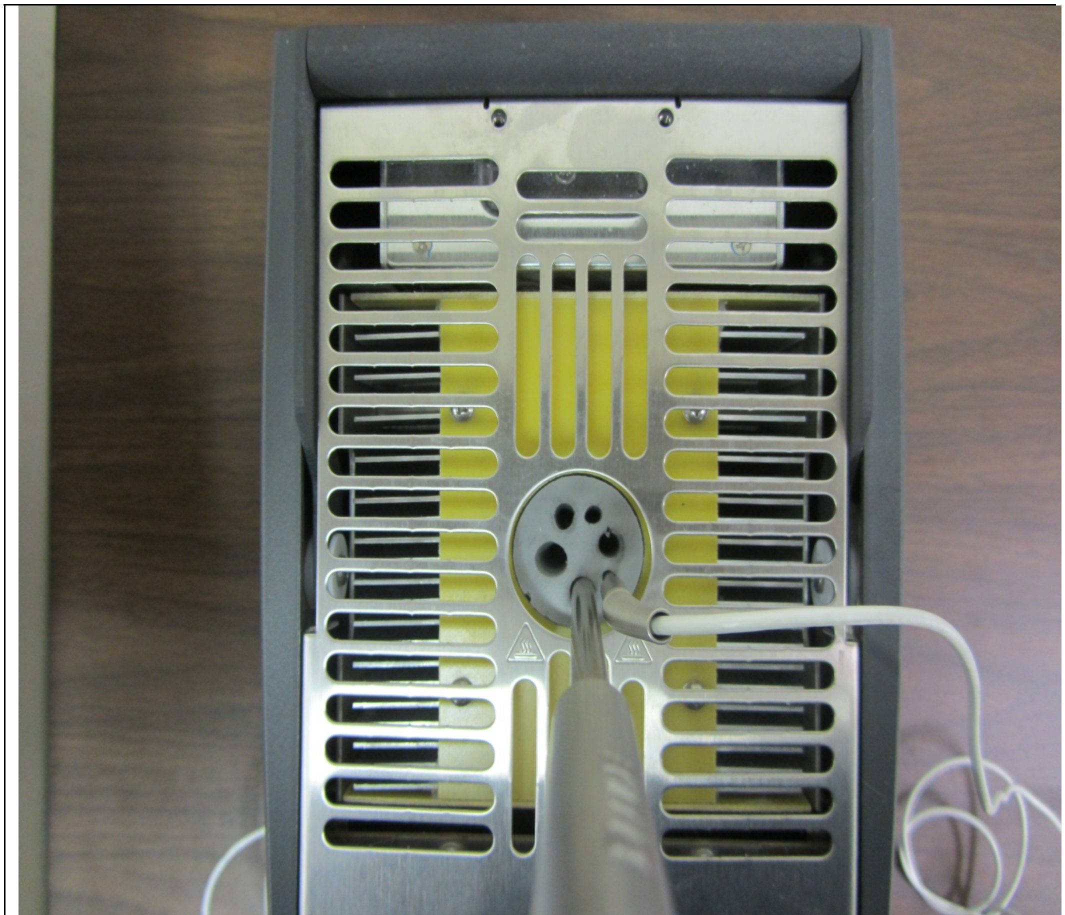


Figure 3: Client's Temp. Probe Inserted

11.4 Power up the candidate temperature standard. Allow a 15 minute warm-up before starting the test.

11.5 Locate the HP laptop computer on the table across from the Fluke reference temperature standard. Double-click on the "CalReport" icon. See Figure 4.

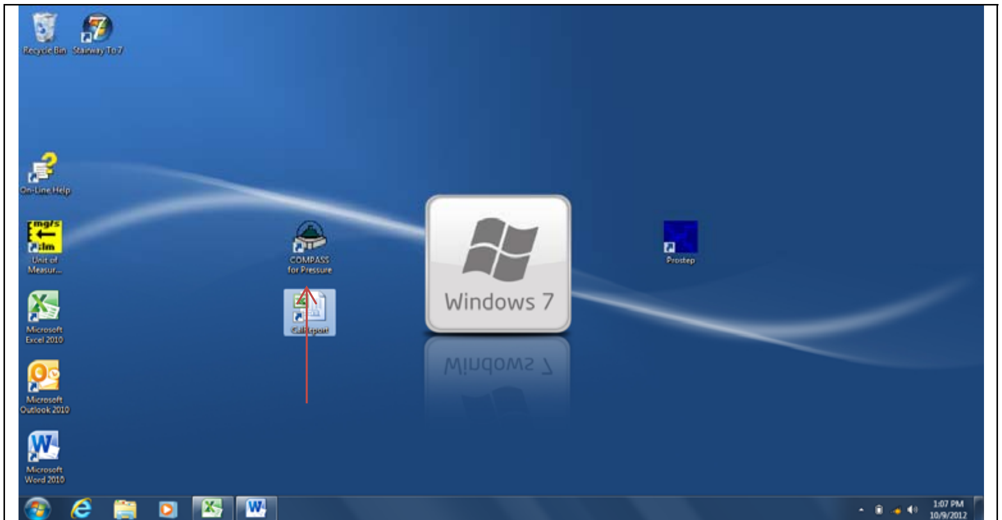


Figure 4: "CalReport" Workbook Icon

11.6 Click on the "TemperatureProbeTest" spreadsheet. Fill in the blue shaded cells with the following information: Room Temperature, Room Pressure, and %RH. This information is located on the Vaisala all-in-one meter located at the top of the gas analyzer instrument rack. Enter the Log number, Customer, Description, Barcode, Serial #, Manufacturer, and Calibration Date. See Figure 5.

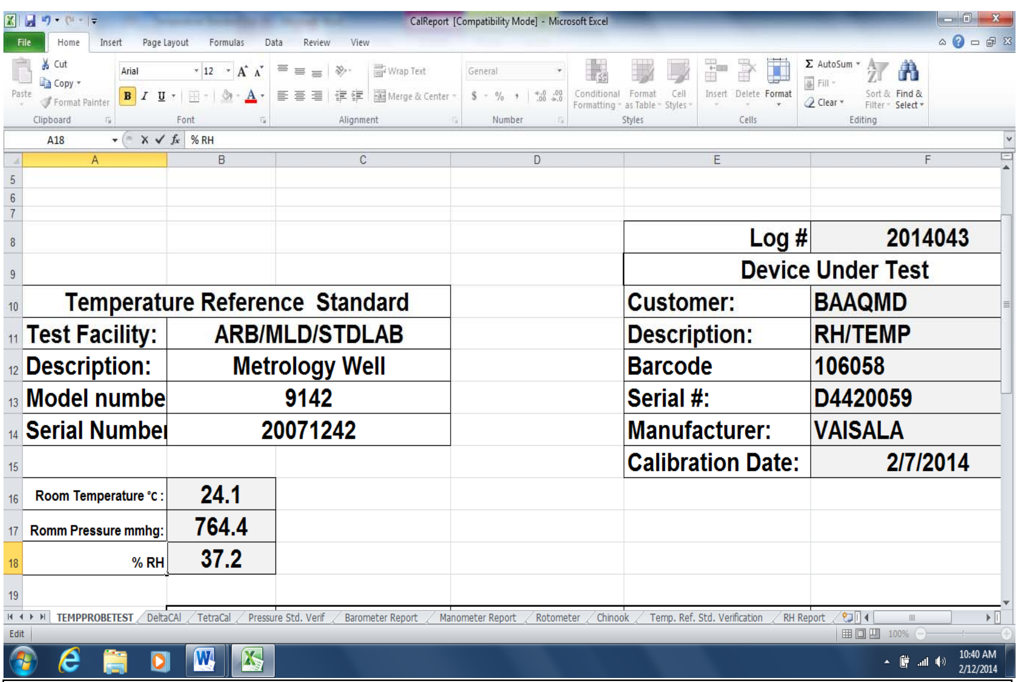


Figure 5: "TemperatureProbeTest" Spreadsheet

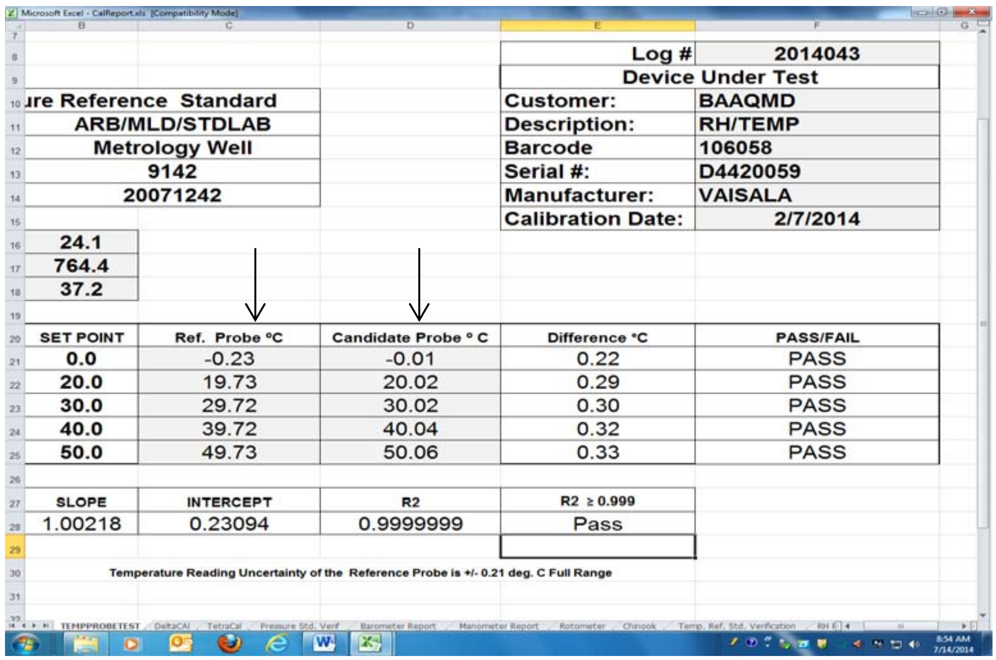
11.7 Press the "SET PT." button on the front panel of the metrology well; then press the enter button. See Figure 6.

Figure 6: Set Point Buttons

11.8 Use the arrow buttons to set the temperature to 0.00 °C; then press the "ENTER" button 2 times. The metrology well will cool to 0.00 °C. There will be an audible beep when the temperature set point has been reached. See Figure 7.

Figure 7: Temperature Set Point Arrows

11.9 Record the "Ref. Probe" temperature reading and the "Candidate Probe" temperature reading onto the spreadsheet. See Figure 8. For the location of reference temperature reading, see Figure 9.



ure Reference Standard	
ARB/MLD/STDLAB	
Metrology Well	
9142	
20071242	

Log #	2014043
Device Under Test	
Customer:	BAAQMD
Description:	RH/TEMP
Barcode	106058
Serial #:	D4420059
Manufacturer:	VAISALA
Calibration Date:	2/7/2014

24.1
764.4
37.2

SET POINT	Ref. Probe °C	Candidate Probe ° C	Difference °C	PASS/FAIL
0.0	-0.23	-0.01	0.22	PASS
20.0	19.73	20.02	0.29	PASS
30.0	29.72	30.02	0.30	PASS
40.0	39.72	40.04	0.32	PASS
50.0	49.73	50.06	0.33	PASS

SLOPE	INTERCEPT	R2	R2 ≥ 0.999
1.00218	0.23094	0.9999999	Pass

Temperature Reading Uncertainty of the Reference Probe is +/- 0.21 deg. C Full Range

Figure 8: Reference and Candidate Probe Temperature Readings

Figure 9: Reference Temperature Reading

11.10 Repeat steps 11.7 through 11.09 at 20.00 °C Set Point.

11.11 Repeat steps 11.7 through 11.09 at 30.00 °C Set Point.

11.12 Repeat steps 11.7 through 11.09 at 40.00 °C Set Point.

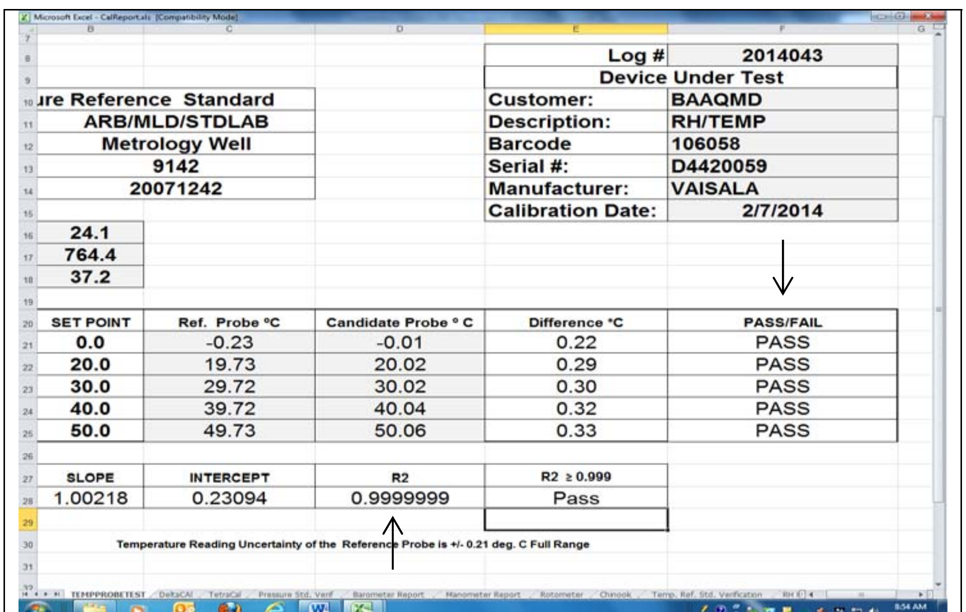
11.13 Repeat steps 11.7 through 11.09 at 50.00 °C Set Point.

11.14 Press the “SET PT.” button then press “F4” to save and end the test.

11.15 Turn off power to the Fluke Metrology Well and the candidate temperature device.

11.16 Carefully remove the temperature probes one at a time from the well.

11.17 If the candidate temperature reading differs by more than ± 0.5°C at any point as indicated in the Pass/Fail column (Figure 10), perform a re-test to confirm the failure. If confirmed, inform the client that the temperature device did not meet criteria and proceed with a SFN (Standard Failure Notification). The SFN is found in the following location: W:\Standards Laboratory\Working Tables. Otherwise, make sure the spreadsheet is accurate and complete.



Log #	2014043
Device Under Test	
Customer:	BAAQMD
Description:	RH/TEMP
Barcode	106058
Serial #:	D4420059
Manufacturer:	VAISALA
Calibration Date:	2/7/2014

ure Reference Standard
ARB/MLD/STDLAB
Metrology Well
9142
20071242

24.1
764.4
37.2

SET POINT	Ref. Probe °C	Candidate Probe ° C	Difference °C	PASS/FAIL
0.0	-0.23	-0.01	0.22	PASS
20.0	19.73	20.02	0.29	PASS
30.0	29.72	30.02	0.30	PASS
40.0	39.72	40.04	0.32	PASS
50.0	49.73	50.06	0.33	PASS

SLOPE	INTERCEPT	R2	R2 ≥ 0.999
1.00218	0.23094	0.9999999	Pass

Temperature Reading Uncertainty of the Reference Probe is +/- 0.21 deg. C Full Range

Figure 10: Temperature Probe Test Spreadsheet

11.18 Print one copy of the spreadsheet and save and close the "CalReport" workbook. Place this copy inside the client's folder.

12.0 Calibration Report Generation

12.1 Double click on the "DASPS" icon. You can find the DASPS icon on any computer located in the Standards Laboratory. See Figure 11.

Figure 11: DASPS icon

12.2 Double click on the "Standards Laboratory" folder. See Figure 12.

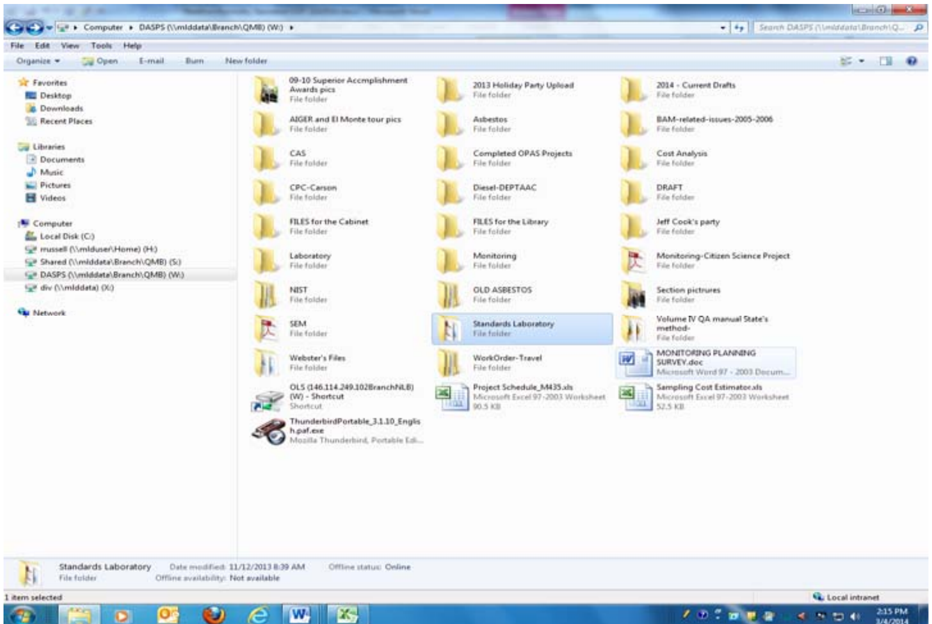


Figure 12: "Standards Laboratory"

12.3 Double click on the "Calibration Forms" folder. See Figure 13.

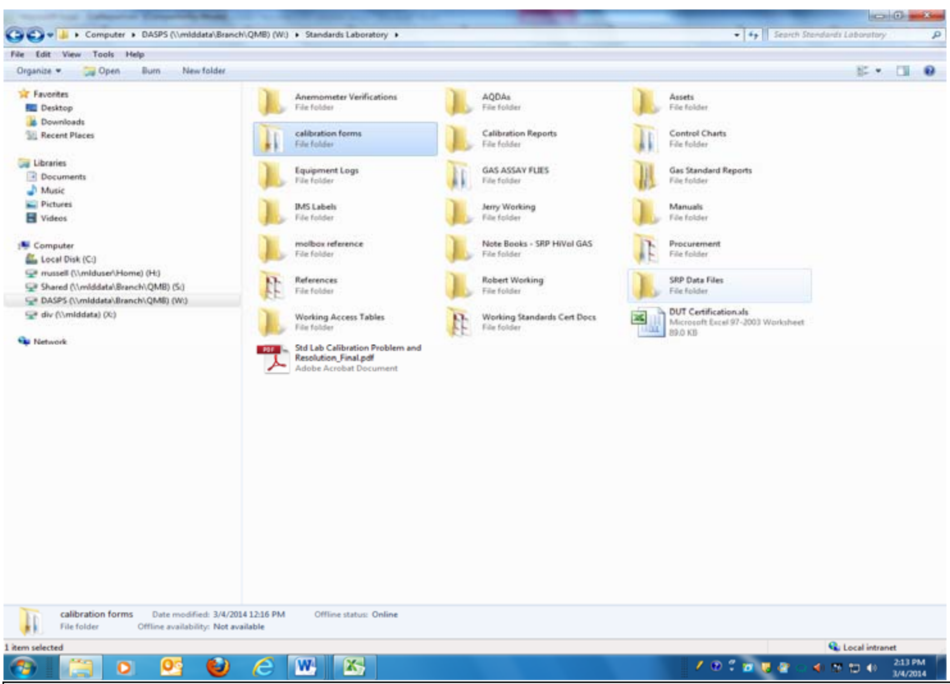


Figure 13: "Calibration Forms"

12.4 Double click on the “Misc Cals” folder. See Figure 14.

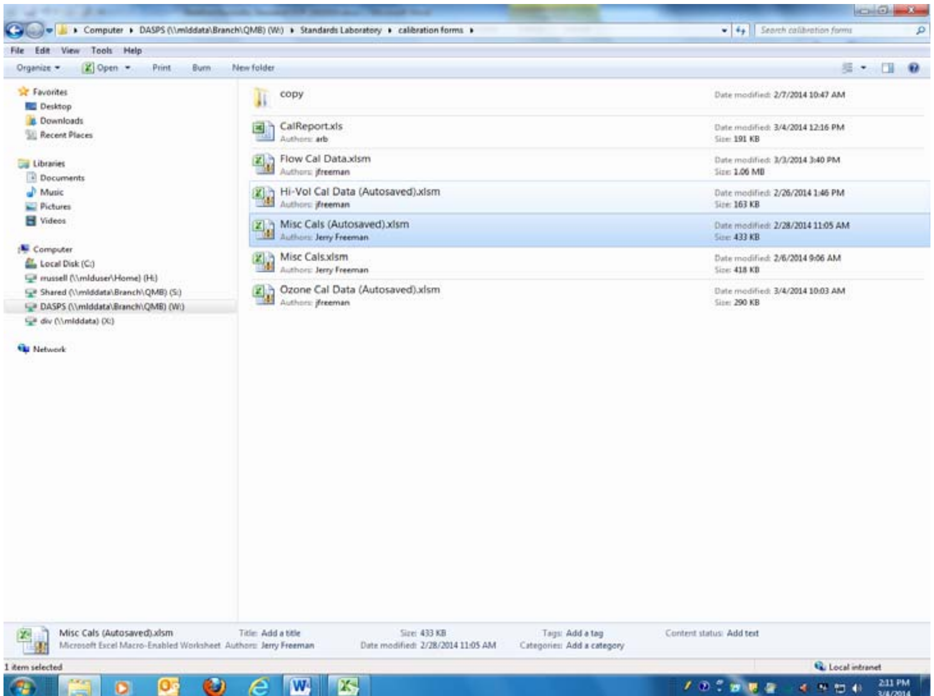


Figure 14: “Misc Cals”

12.5 Click on the “Login” worksheet. See Figure 15.

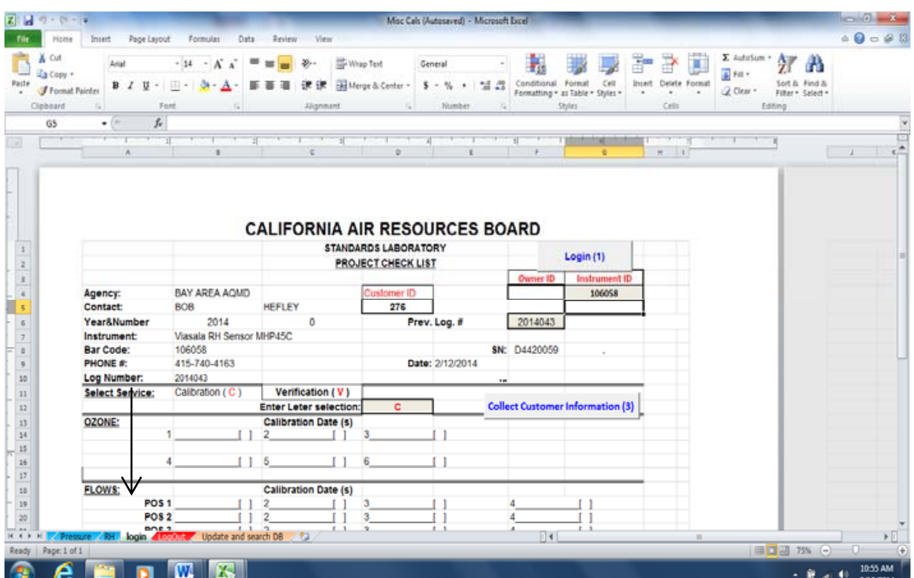


Figure 15: “Login” worksheet

12.6 Enter instrument ID. See Figure 16.

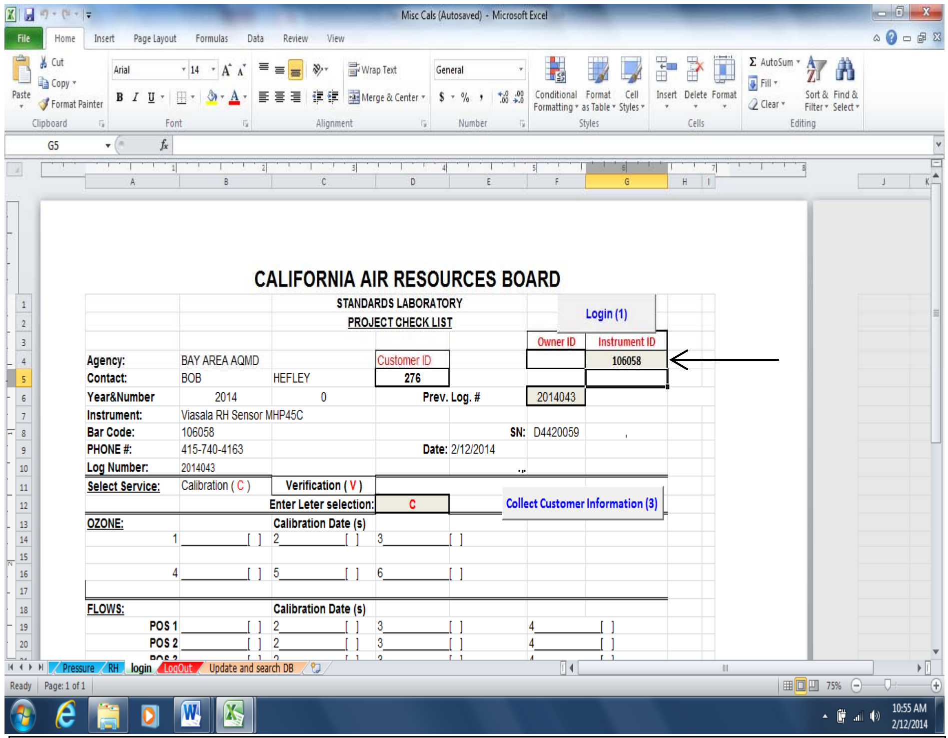


Figure 16: Instrument barcode

12.7 Click on the “Collect Customer Information” button. See Figure 17.

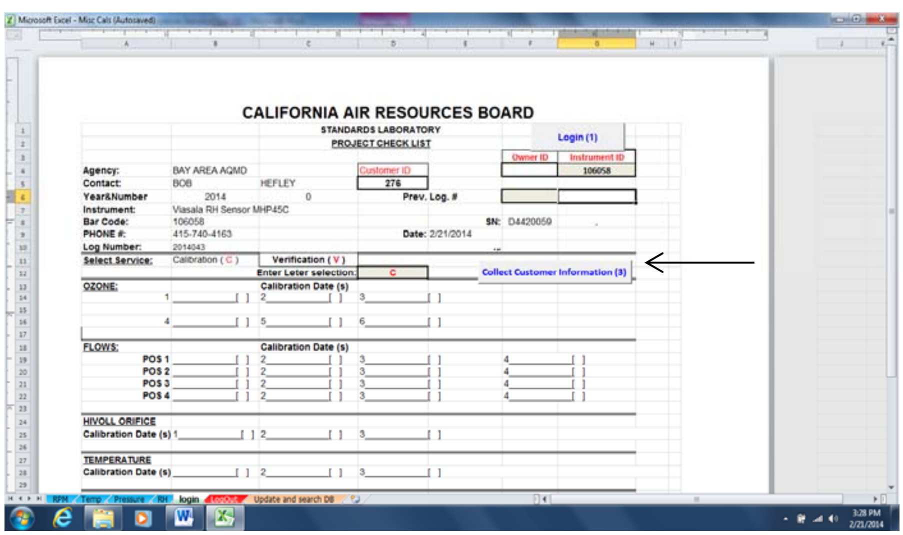


Figure 17: “Collect Customer information”

12.8 Click on the “Temp” worksheet. See Figure 18.

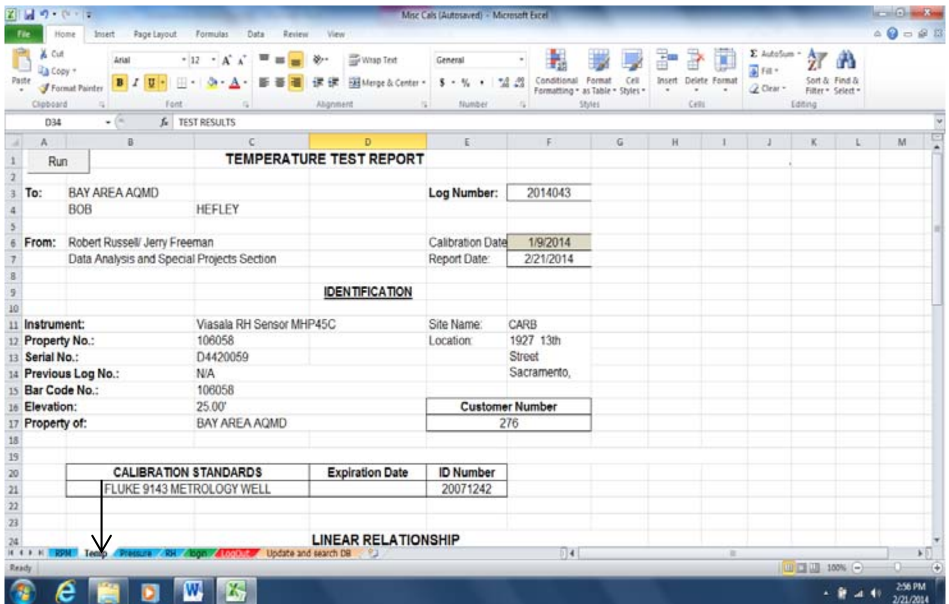


Figure 18: “Temp” worksheet

12.9 Transfer the temperature readings in the “Ref Probe” column (Figure 10) to the “Reference Output” column (Figure 19). Transfer the temperature readings in the “Candidate Probe” column (Figure 10) to the “Candidate Output” column (Figure 19).

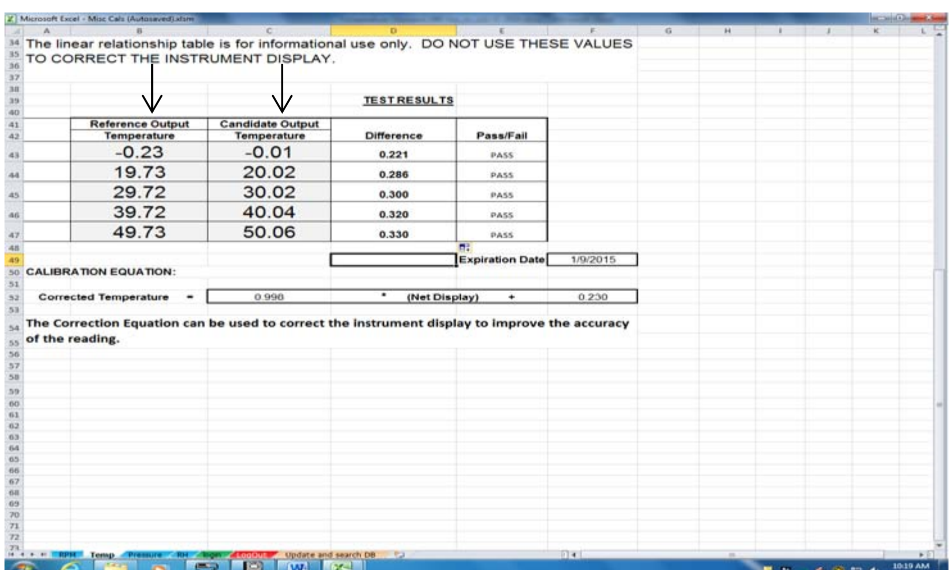


Figure 19: “Temperature Test Report” worksheet

12.10 Click on the “RUN” button once to save data. See Figure 20.

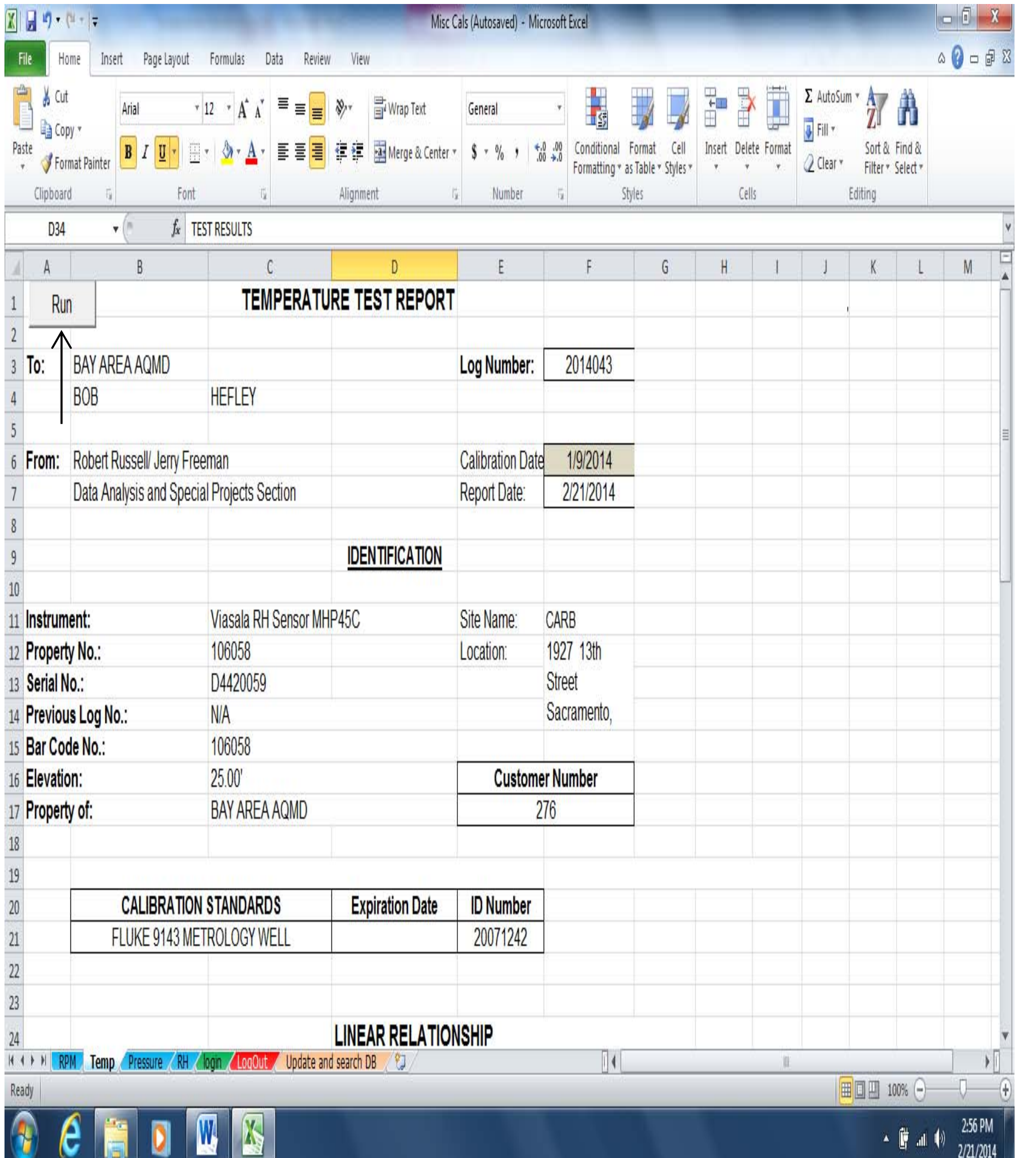


Figure 20: “Temperature Test Report”

12.11 Have a peer review the “Temperature Test Report” for completeness and accuracy. Print two copies of the test report. Place one copy with the client’s temperature device and file the other copy along with the “CalReport” spreadsheet inside the customer’s login folder. Convert the Temperature Test Report to a PDF document. Save in the following location: W:\Standards Laboratory\Calibration Reports.

13.0 Quality Control

- A semi-annual multipoint comparison is performed between the Fluke temperature reference standard and the Omega HH41 Digital Thermometer. The acceptance criteria are ±0.5 °C at each point and an R2 ≥ 0.9999. If all criteria are not met, re-run the test to confirm the result. If confirmed, consult with lead staff to determine if the problem is with the control standard or reference standard. If it is determined to be an issue with the control standard or the reference standard, have the device sent to a qualified vendor that utilizes NIST traceable standards for repair or re-calibration. If the test meets all criteria, the slope and intercept is added to the temperature control chart located in the following location: W:\Standards Laboratory\Control Charts.

14.0 Troubleshooting

- If the reference or candidate temperature device does not reach the set point within ±0.5 °C, check to see if the probe was inserted according to section 11.2.

- If the temperature well does not operate according to this operation procedure, refer to the Troubleshooting Guide on page 101 of the "Field Metrology Well Technical Guide."

15.0 References

- Model 9142 Fluke Field Metrology Well Users Guide.

- U.S. EPA Quality Assurance Handbook for Air Pollution Measurement Systems Volume IV: Meteorological Measurements. Version 2.0.